

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of Bachelor of Technology (B.Tech)

University Examination May 2011

CE215 Object Oriented Programming

Date: 21.5.2011, Saturday

Time: 10:00 a.m to 1:00 p.m

Maximum Marks:70

Instructions:

1. Attempt all questions.
2. Answers both sections in separate answer books.
3. Figure to the right indicates full marks.
4. Assume suitable data, if necessary and mention it.

SECTION – I

Q.1(a) Write down the appropriate answer from the given options. 05

1. What is the correct value to return to the operating system upon the successful completion of a program in C++?
A. -1 B. 1 C. 0 D. Programs don't return a value.
2. Which of the following is not a correct variable type?
A. float B. real C. int D. double
3. Which of the following is not a looping statement in C?
A. while B. until C. do D. for
4. Which operator from the list you can't overload?
A. + B. <> C. sizeof D. *
5. Functions can be declared with default values in parameters. We use default keyword to specify the value of such parameters.
A. True B. False

Q.1(b) Answer the following questions. 06

1. Explain Switch...Case statement with example.
2. Write a simple code to explain function overloading.

Q.2 Answer the following questions. 12

1. Write a function in C++ to find the area of a triangle. Also write main to test the function.
2. Write a short note on Operators of C++.

OR

Q.2 Answer the following questions. 12

1. Write a program to find out the entered number is perfect or not. 06
2. List out the rules of Virtual function. 04
3. Define Static binding & late binding. 02

Q.3 Answer the following questions. 12

1. What is the application of scope Resolution Operator in C++? And Explain setw and endl manipulators
2. Short Note: Benefits of OOP
3. Explain different looping statement of C++ with example.

OR

Q.3 Answer the following questions. 12

1. Explain Inline Function with example.
2. Write a code to explain Constructor concept.
3. Explain structure of C++ Program.

SECTION – II

Q.4 Do as directed. 11

1. Draw MATLAB's schematic diagram with main features. 03

2. Explain the following commands in MATLAB. 04
(a) close (b) linspace (c) plot (d) clc
3. Write a computer graphics program to draw a circle in the center of the computer screen having radius 250 units. 04

Q.5 Answer the following questions.

12

1. Draw Event Trace Diagram for ATM scenario.
2. Define: Computer Graphics. Also explain its components.
3. What is the purpose of models in designing? What do you mean by views and stages of a model?

OR

Q.5 Answer the following questions.

12

1. Draw state chart diagram for phone line.
2. Explain Event and State in detail.
3. Explain the uses of discriminator with an example.

Q.6 Answer the following questions.

12

1. Define: metadata. How is it different from patterns?
2. Differentiate:
 - a) Query and Operation
 - b) Activity and Operation
3. Can Aggregation be fixed, variable or recursive? Why?

OR

Q.6 Answer the following questions.

12

1. Write down two examples of derived attributes? Why are you considering them as derived attributes?
2. Explain: Layered approach in software engineering.
3. What kind of control information represented by dynamic model? And what kind of models does OMT contain?
